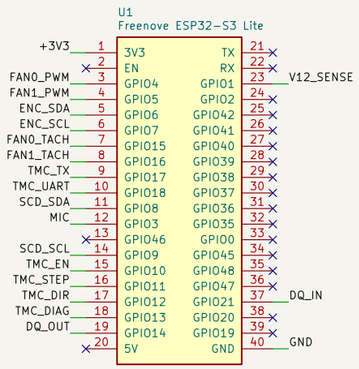
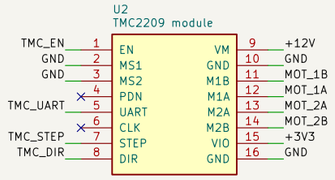
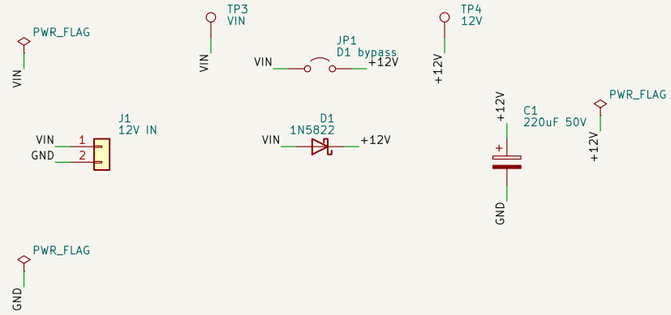
## ESP32-S3 (Freenove Lite, FNK0099), soldered: USB toward the box's left wall



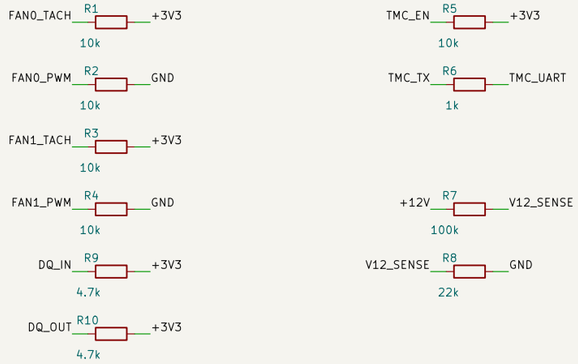

## TMC2209 stepper driver (StepStick module), soldered



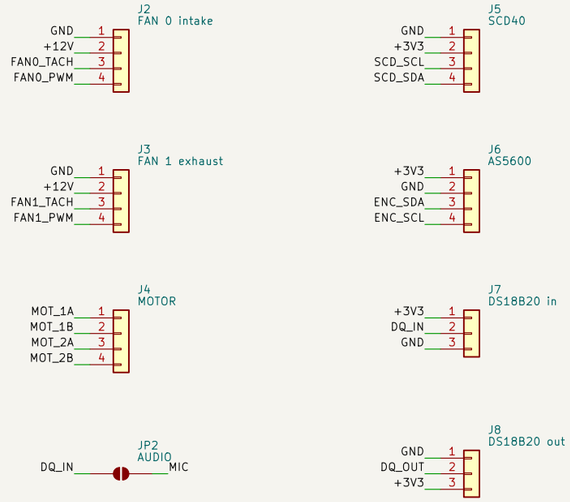

## 12 V input: D1 against reverse polarity; a jumper on JP1 bypasses it




## Fan, temperature-probe and driver resistors; 12 V sense divider




## Connectors (2.54 mm male headers)




## Test pads




## Mounting holes (M2)




Nets with the same label are connected. Pins as in firmware/libs/WindowVent/src/Pins.h.
U1 pins 1–20 are its left header (3V3 … 5V) and 21–40 its right header (TX … GND), top to bottom as on the board.
U2: MS1 = MS2 = GND (UART address 0); CLK and PDN free. DIAG is not on this module's pins: GPIO 13 goes to TP5.
R6 1k: single-wire UART, TX through 1k and RX straight onto PDN_UART. R5 holds the driver off (EN high) while the ESP32 boots.
R7/R8: 12 V reads 2.16 V on GPIO 1 (ADC1). R2/R4 keep the fans slow at boot; R1/R3 tach pull-ups; R9/R10 1-Wire pull-ups.
GPIO 38, 39, 40 stay free for the INMP441 microphone reserved in Pins.h (no room for its header on this board).
AS5600: tie its DIR pin to GND at the module. C1 sits next to U2's VM pin.
JP2 (solder jumper on the back, open): bridge it for the audio mode, a microphone on J7 read on GPIO 3 (GPIO 21 has no ADC); leave R9 out then.

Generated by tools/pcb_schematic.py from tools/pcb_parts.py
github.com/mmassico/window_air_vent

Sheet: /
File: window_vent_ctrl.kicad_sch

**Title: Window vent controller**

| | | |
|---|---|---|
| Size: A3 | Date: 2026-11-02 | **Rev: 1.1** |
| KiCad E.D.A. 10.0.7 | | Id: 1/1 |